DeepLoRa: Learning Accurate Path Loss Model for Long Distance Links in LPWAN

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***Abstract*—LoRa (Long Range) is an emerging wireless technology that enables long-distance communication and keeps low power consumption. Therefore, LoRa plays a more and more important role in Low-Power Wide-Area Networks (LPWANs), which easily extend many large-scale Internet of Things (IoT) applications in diverse scenarios (e.g., industry, agriculture, city). In lots of environments where various types of land-covers usually exist, it is challenging to precisely predict a LoRa link's path loss. As a result, how to deploy LoRa gateways to ensure reliable coverage and develop precise fingerprint-based localization becomes a difficult issue in practice. In this paper, we propose DeepLoRa, a deep learning-based approach to accurately estimate the path loss of long-distance links in complex environments. Specifically, DeepLoRa relies on remote sensing to automatically recognize land-cover types along a LoRa link. Then, DeepLoRa utilizes Bi-LSTM (Bidirectional Long Short Term Memory) to develop a land-cover aware path loss model. We implement DeepLoRa and use the data gathered from a real LoRaWAN deployment on campus to evaluate its performance extensively in terms of estimation accuracy and model transferability. The results show that DeepLoRa reduces the estimation error to less than 4 dB, which is 2× smaller than state-of-the-art models.**

***Index Terms*—Wireless networks, Low power wide area networks, Attenuation measurement, Propagation losses, Recurrent neural networks.**

I. Introduction

The development of Internet of Things (IoT) has witnessed the explosion of applications, IoT devices, and network size. In many scenarios (e.g., agriculture, industry, city, home, health-care), a large amount of unattended IoT devices are deployed, sending a small volume of data sporadically, which are expected to last for years given limited energy. To simultaneously fulfill all these requirements, extending short-range and low-power wireless techniques with ad-hoc architecture (e.g., wireless sensor networks [1]–[3]) is a common solution in the past decades, which suffers from dramatically increasing deployment and maintenance cost with the increase of network scale. To mitigate this gap, wireless techniques (e.g., LoRa, Sigfox, NB-IoT) for Low-Power Wide Area Networks (LPWANs) have recently emerged. Due to low-cost COTS radio/gateway (e.g., Semtech) and open-source development (e.g., LoRa Alliance), LoRa is gaining popularity in both industry and academy areas [4]–[6].

LoRa physical layer adopts chirp spread spectrum (CSS) modulation [7] to enable data packet reception under low signal-to-noise ratio (SNR) (e.g., -20dBm) while keeping low power consumption (e.g., 400mW transmitting at 20dBm, 5μW in idle mode) as its low duty cycle and narrow bandwidth. The high signal sensitivity enables COTS LoRa radio and gateway to receive the potential weak signals. Hence, LoRa obtains a large link budget [4], [8], [9], which accounts for its high maximum feasible power loss along with the signal propagation between an end node and a gateway. The sufficient link budget can provide reliable coverage spanning from several kilometers to tens of kilometers in various environments (e.g., urban area, rural area).

Though LoRa enables long-distance links, we observe that the communication distance may vary greatly in real-world deployments. When an end node is deployed at different directions of a gateway, the power attenuation of the link, called *path loss*, changes due to different types of land-covers (e.g., tree, buildings, road) along the path. An accurate path loss model is vitally important for LoRaWAN applications. Since path loss correlates to the packet delivery probability of a link [10], if we can accurately predict the path loss associated with a LoRa gateway before it is deployed, we can optimize the LoRaWAN coverage by selecting gateway locations. Moreover, in LoRaWAN, end node localization [11]–[14] relies on the matching of the signal fingerprint (e.g., received signal strength indicator (RSSI)) observed by several gateways. If we can accurately predict path loss without an exhausted site survey, the localization system will be deployed and maintained with low overhead. However, facing the environment diversity in different and large areas of LoRaWAN, it is challenging to develop such an accurate and general path loss model with low overhead [14], [15].

In this paper, we propose DeepLoRa, a learning framework for accurate path loss estimation of long-distance LoRa links. We have two key observations. First, public remote sensing images [14], [15] can be utilized to recognize the fine-grained land-covers distributed along a link. Second, the influence of the land-covers on path loss is actually very complicated. Both the types of land-covers and the order they appear along the link make a difference (Section III-B). Then, we resort to deep learning technique [16]–[18] to model the influence of a specific land-cover distribution on path loss.

Specifically, instead of considering the environment of a LoRa link as a whole, DeepLoRa divides it into an ordered sequence of short links (called *micro link*) with the same length. The detailed land-covers of each micro link are recognized by utilizing remote sensing images. Then we apply a deep neural network with Bidirectional Long-Short-Term-Memory

(Bi-LSTM) units for sequence analysis to learn a path loss model based on the measurements collected from the area of interest. Our Bi-LSTM model inherently models the mapping from land-cover types and order to path loss. When we have trained our model, with only a few extra data collection and model finetuning, the model can be directly transferred to areas with similar land-cover composition.

We implement DeepLoRa and evaluate its performance on the dataset collected from a campus LoRaWAN deployment spanning $6 \times 6km^2$ area in an urban scenario. The dataset involves more than 30,000 packets logged by two gateways sent by 6 mobile end nodes. Experiment results show that DeepLoRa achieves a mean error of 3.56dBm, which is $2\times$ smaller than state-of-the-art models. When we transfer the model from one gateway to the other and finetune it with less than 200 data records, it achieves a mean error of 4.79dBm. Our contributions are summarized as follows:

1) Instead of the physical model, we first propose to utilize deep learning for path loss estimation of long-distance LoRa links across a large area in outdoor scenarios.
2) We empirically study the influence of detailed land-cover sequence on path loss in a real LoRaWAN system. We propose DeepLoRa utilizing adaptive Bi-LSTM model to study the relationship between path loss and the corresponding types and order of land-covers.
3) We implement DeepLoRa and evaluate its performance in real LoRaWAN deployment. The experimental results show that the mean error of DeepLoRa is $2\times$ smaller than state-of-the-art models, and our model can be generalized with low overhead.

The rest of the paper is organized as follows. We introduce the related work in Section II. In Section III, we introduce the problem background and our motivation by illustrating environment modeling challenges. The system design of DeepLoRa is followed in Section IV. Section V and Section VI exhibit the implementation and evaluation, respectively. We conclude our paper in Section VII.

II. Related Work

The characteristics of long-distance wireless links have been empirically studied and theoretically modeled in the past decades. We summarize the existing efforts from the following three aspects.

LoRaWAN field studies: LoSee [10] shows two gateways are needed to ensure the full coverage of the $4.5km^2$ campus. In the deployment of Liando et al. [19], the maximum line-of-sight (LOS) and non-line-of-sight (NLOS) communication distances are approximate 9.08km and 2km when the packet delivery ratio is higher than 70%. Numerous other empirical studies [4], [9], [20]–[24] have been conducted to measure LoRa coverage ranges. The specific range varies with experiment environments. For example, Centenaro et al. [23] observe the range of 2km in an area of high-buildings. Wixted et al. [9] observe coverage of 1km to 20km in the central business district. Different conclusions on LoRa coverage by these empirical studies reveal that path loss increases with communication distance at different rates in various environments. How to model the impact of the environment on path loss is the focus of designing path loss models. In DeepLoRa, we deploy 2 gateways and 6 mobile end nodes in a campus environment to study the detailed relationship between the land-covers along a LoRa link and the path loss.

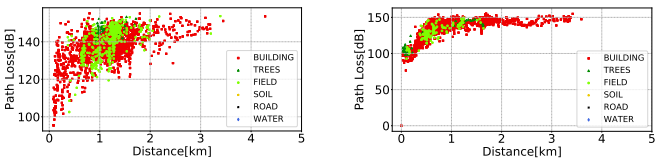
Land-cover and environment aware models: A few models are integrating environmental factors to reflect the difference in the rates of path loss increasing. Empirical model Okumura-Hata [25], [26] provides ready-to-use formulas suitable for different environments (e.g., urban, suburban, rural areas). Since those formulas were fitted using data collected in Tokyo, Japan, directly applying this model in new environments with different changing rules of path loss would lead to inaccurate predictions. Bor et al. [4] adopts the well known log-normal shadowing path loss model [27]. Unlike free-space path loss, Bor model uses on-site measurements to estimate the absolute path loss by path loss exponent. These two models only adopt regional environment information for prediction. In this case, the same deployment area would result in the same formulation or path loss exponent, while land-cover compositions of LoRa links are anisotropic. Lacking per-link knowledge of the environment would cause obvious ambiguity in path loss estimation. Demetri et al. [15] and SateLoc [14] adopt remote sensing techniques to analyse the composition of land-covers along LoRa links quantitatively. With the types of land-covers along a LoRa link, Demetri et al. [15] decide which Okumura-Hata formula to use based on the dominating land-cover type. So the effect of land-covers is not directly reflected in the prediction. SateLoc [14] divide the whole link into segments by different land-covers and calculate the overall path loss segment by segment using Bor model with the corresponding path loss exponent, but the result would be of no difference if the order of those segments changes in the link. ALL models mentioned above do not fully utilize the fine-grained environment information. Also, they can not well transfer to new environments due to fixed environment modeling.

Machine learning based models: Some works [28]–[32] use machine learning to model path loss. Oroza et al. [28] adopt random forest algorithm to predict the path loss for American River Hydrologic Observatory wireless links. Zhang et al. [29] use random forest and KNN for evaluating the unmanned aerial vehicle communication channels. Cheng et al. [31] associate the floor plan of a building to RSSI values in each indoor Wi-Fi measurement by Convolutional Neural Networks (CNNs). According to our best knowledge, there is no learning-based approaches for outdoor LoRa path loss estimation yet.

Instead of adopting the physical path loss model, we adopt Recurrent Neural Network based model to depict the complex relationship between the path loss and the types and order of land-cover along the path for more accurate path loss estimation. Furthermore, our model are more transferable because of using less-information-lost raw environment data and highly generalizable RNN models.

TABLE I
THE TYPES OF LAND-COVERS.

NLOS	BUILDING	buidings
	GREENHOUSE	greenhouse structures
	TREES	trees
LOS	Field	farming field or glassland
	SOIL	bare soil
	ROAD	streets,roads and highways
	WATER	lakes and rivers



(a) G_1 (b) G_2

Fig. 1. Path loss vs. distance for different land-cover dominated links regarding to different gateways G_1 and G_2.

III. PROBLEM BACKGROUND AND MOTIVATION

A. LoRa Path Loss Estimation

Generally, actual signal path loss can be calculated by :

$$PL = P^t + G^r + G^t - P^r \tag{1}$$

where P^t is the power fed into the transmitter's antenna, P^r is the Received Signal Strength Indicator(RSSI), G^r and G^t are the power gains at receiver and transmitter sides, respectively. Those terms are in dB(m) units.

In practice, given a received LoRa packet, we can obtain its RSSI and SNR from the gateway, but RSSI is the power combination of LoRa signal and various noises. To eliminate the influence of the noises, we use Expected Signal Power (ESP) [15] as a metric to indicate actual received signal power, which can be derived from the following equation:

$$ESP = RSSI + SNR - 10 \cdot \log_{10}(1 + 10^{0.1SNR}) \tag{2}$$

SNR is in dB unit, the other terms are in dBm units.

Among the models of signal path loss, free-space path loss (FSPL) derived from Friis transmission equation [33] models the path loss in ideal free space scenario which is unobstructed and not affected by multi-path effect, which is given by:

$$FSPL(d) = 10\log_{10}(d) + 20\log_{10}(f) - 27.55 \tag{3}$$

where d is the distance between transmitter and receiver whose unit is m, f is the frequency whose unit is MHz.

Researchers adopt various ways to measure the path loss of LoRa links in large open spaces with few obstacles to simulate free-space condition, e.g., using High-altitude weather balloon [15], conduct experiments on beaches/sea with clear sights [19], [20]. The results showed that even in a carefully selected LOS environment, the error between the actual path loss with the path loss estimated by FSPL is still appreciable.

B. The Challenge Of Environment Modeling

As free space conditions can hardly be achieved in reality and FSPL only provides a rough lower bound of the path loss estimation in real LoRa deployment, real-world path loss of LoRa links is susceptible to attenuation caused by the environment.

Environment aware models require acquiring environmental information. Original physical models like Okumura-Hata model and Bor model mainly adopt two approaches: Naked-eye estimations based on experience or on-site measurements, which are usually labor exhausted, especially in long-distance scenarios. Both approaches can only provide rough regional environment information. If two LoRa links are in the same deployment area but traverse different kinds of land-covers, they still could be modeled in the same environment using the same formulas/path loss exponents.

We conduct an empirical study based on our LoRaWan system (introduced in IV-D) measurements to find rules according to which different land-covers affect the path loss. All the measurements are collected in the same area. As shown in Figure 1, we color the measured points according to their dominating land-covers along the link. We can see different links have different dominating land-covers (e.g., BUILDING, TREES and FIELD as defined in Table I). Moreover, even under the same distance, the path loss distribution varies for different types of dominating land-covers. BUILDING and FIELD make the path loss more dynamic than other types of land-covers. The result not only shows that different types of land-covers will lead to diverse effects on the path loss, but also emphasize the heterogeneity of LoRa links in the same area. So we need to elaborate per-link environment information for accurate LoRa path loss estimation.

Remote Sensing based Land-cover Recognition. Using remote sensing frees researchers from tedious measurement work while providing rich and fine-grained knowledge of the environment. Literally, remote sensing acquires multi-spectral images of the large-scale area on the earth remotely using aircrafts or satellites equipped with sensors that detect radiation reflected or emitted from target objects. By extracting features from those images and classify them with machine learning models(e.g., Support Vector Machines, Random forest), we can recognize different types of land-covers. In our case, we consider several typical land-covers and divide them into two groups according to whether they may lead to NLOS signal attenuation or not (i.e., LOS transmission) as shown in Table I.

For the two models that utilize remote sensing with physical models, they extend from regional environment modeling to per-link modeling. However, [15] neglects the effect of detailed land-cover types along the link. Because instead of using naked eyes to decide which Okumura-Hata formula to use, [15] utilizes the dominating land-cover type. They choose to use suburban formula if dominating type belongs to LoS category and use urban formula otherwise. Okumura-Hata model actually becomes bottleneck of the whole path loss estimation pipeline. SateLoc [14] ignores the order that different land-covers appear in the link. Because they just divide the whole link into segments and add up the path loss